



Paper ID : 260812

Printed Page: 1 of 1

Subject Code:MPC102T

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MPHARM

(SEM. I) THEORY EXAMINATION 2025-26

ADVANCED ORGANIC CHEMISTRY -I

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x10 = 20

- What are Carbenes? Give one method of their formation.
- Define Hoffman's Rule of elimination with an example.
- State the synthetic utility of Ozonolysis.
- Write the structure and use of N-bromosuccinimide (NBS).
- What is the role of Wilkinson's Catalyst in organic synthesis?
- Write down the mannich reaction.
- What do you mean by Functional Group Interconversion (FGI)?
- Write down the uses of the Wilkinson reagent.
- Why is the protection of the Amino group necessary during peptide synthesis?
- Give the reaction for Sandmeyer Reaction.

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- Discuss the mechanism, orientation, and stereochemistry of S_N1 and S_N2 reactions. Compare them based on solvent and substrate effects.
- Explain the Retrosynthetic approach (Disconnection approach). Describe the guidelines for the dissection of molecules and explain C-X disconnections with examples.
- Explain in detail reaction, reaction mechanism, and synthetic applications of the Sharpless asymmetric epoxidation reaction.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- Explain the formation, stability, and synthetic applications of Free Radicals.
- Describe the mechanism and applications of Baeyer-Villiger Oxidation.
- Discuss various ways to determine reaction mechanisms.
- Discuss the synthesis of ketoconazole and alprazolam.
- Explain in detail the principles, terminologies, and advantages of retrosynthesis.
- Describe in detail the C-C disconnections.
- Explain the mechanism and synthetic applications of Wittig reaction.